From Land to Liquidity: Land Acquisition Compensation and Elite Mobility in Taiwan's Land Reform

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Data and Measurement

Short-run baseline population results

Estimating and controlling confounding factors from the land reform

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Motivation

- Taiwan's land reform has long been viewed as the root of the "Taiwan Miracle." However, existing studies find little improvement in agricultural output through property rights and only marginal effects through relaxation on restrictions of crop choice.
- Its effects are better understood through the transformation of social structure and the resulting political and economic differentiation.
- Land-to-the-Tiller converted land wealth into liquid assets, facilitating the optimal allocation of people across places and occupations. Different compensation rules for dry and wet land provides exogenous variations for causal interpretations.

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Three stages of postwar land reform

Year	Name	Land or contractual relationship	Administrative action
1949	375 Rent Reduction	Existing Tenant contracts	Rent ceiling, tenure protection, and contract registration
1951	Sale of Public Land	Government-held cultivated land already under tenancy	Sale to incumbent cultivators with installment payment
1953	Land-to-the-Tiller	Privately owned rented land above legal retention limits	Government acquisition and resale to incumbent tenants

Literature: tenancy and agricultural output

Cheung (1969) and Liu (2022)

- Cheung's share-tenancy analysis treats the 37.5% rent rule as a share of realized harvest.
- Liu shows that the ceiling used standard yield. Rent was fixed with respect to realized output.
- The usual sharecropping incentive argument therefore does not describe this rule well.

Kim and Wang (2025)

- They test the direct effect of reform on agricultural output and find little robust gain from the private-land transfer.
- Colonial extension and farmer training had already spread modern techniques. Another yield jump may have been limited.
- Freedom to change crops on former public land may offer a sharper output margin.

Literature: land as a financial and social asset

Koo's recollections

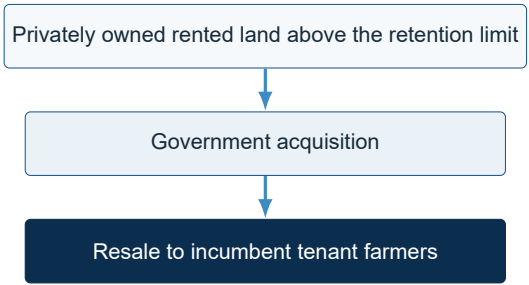
- The account collected in *Jin Han Mei Xiang* emphasizes the release of land wealth into more liquid assets.
- Koo participated closely in reform while standing outside government. His observations provide a distinct participant perspective.

Qu and He

- Qu traces landlord compensation into enterprise shares and industrial finance.
- He reads reform as a change from village-based relations to modern individual property rights and state governance.

This paper studies population reallocation, long-run employment structure, and local electoral outcomes.

The 1953 Land-to-the-Tiller transaction



Compensation to former landlords

- 70% of land value in commodity-denominated land bonds
- 30% in shares of state-owned enterprises
- Bonds paid 4% annual interest and were amortized over ten years
- Outside Penghu, repayment was scheduled twice per year

Statutory text: bond denomination & repayment asset

Original statute: 《制定臺灣省實物土地債券條例》／總統府公報第 383 號／ 1953 年 1 月 26 日

本債券券面實物，分稻穀與甘薯二類：一、稻穀債券，定額十二億六千萬公斤，用以補償徵收田地目部份所包括之兩季田、單季田、看天田、輪作田（包括特殊田）地價。二、甘薯債券，定額四億四千萬公斤，用以補償徵收畑地目部份地價。以上二類債券，由臺灣省政府於民國四十二年度內按照債券面額十足發行。

本債券到期本息，依左列規定償付之：一、稻穀債券，償付稻穀。但用以補償輪作田（包括特殊田）地價所搭發之稻穀債券，按該年期當地稻穀時價折付現金。二、甘藷債券，按該年期之甘藷時價折付現金。前項現金本息，由臺灣土地銀行當地行處經付，其稻穀本息，得轉託糧食局指定之當地倉庫經付，其經付稻穀標準，依臺灣省田賦徵實實收規則之規定。

Both land types: acquisition price = **2.5 times standard annual yield**. 70% in **tax-exempt commodity land bonds**, with **4% annual interest over ten years**.

Dryland	Sweet-potato bonds	Cash repayment at Taiwan Land Bank branches
Paddy land	Paddy bonds	In-kind repayment at designated local warehouses

Bond liquidity and spatial lock-in

Grain procurement and distribution

The Food Bureau organized grain procurement and distribution. Restrictions on private grain transactions limited the liquidity of in-kind paddy repayments.

Emerging securities markets

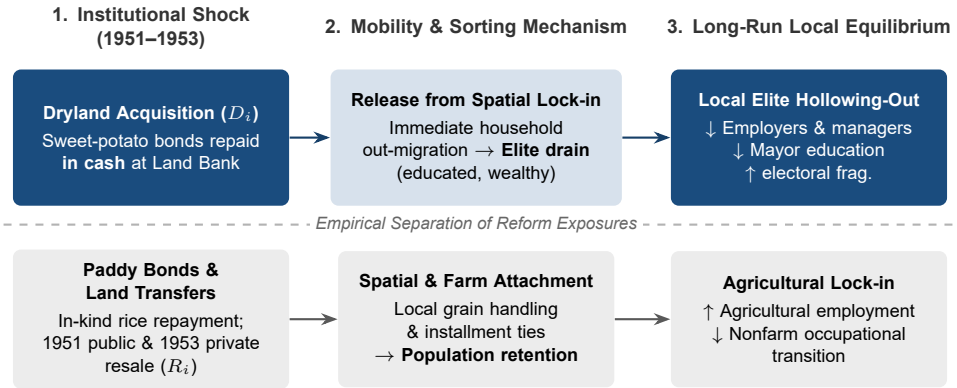
Official and private securities markets began to develop. Cash-paying sweet-potato bonds were easier to discount or pledge in secondary and informal credit markets, at lower transaction costs.

Cash repayment releases the household from spatial lock-in

No local paddy collection or handling for ten years → Immediate relocation of the entire household → Cities or other locations suited to nonfarm work

More liquid compensation facilitates landlords' migration and transition into nonagricultural occupations.

Conceptual framework: from land reform to structural differentiation



Core identification insight: Isolating the cash-compensation channel (D_i) from overall land transfer (R_i) and public-land sales ($PublicLease_i$) unmasks a sharp divergence between **mobile former elites** and **tied land recipients**.

Results in brief

- **Population:** In the baseline DID, a 10 pp higher township dryland share corresponds to about 170 fewer residents after reform.
- **Migration:** More people live and work outside their birth township.
- **Employment:** Fewer employers and managers remain locally.
- **Elections:** More candidates, lower winners' vote shares, and narrower victory margins.
- **KMT support:** Greater support through elite outflow, rather than gratitude toward the authorities or greater conservatism among local residents.
- **Public land:** Land recipients remain in agriculture, with fewer transitions into nonagricultural work.

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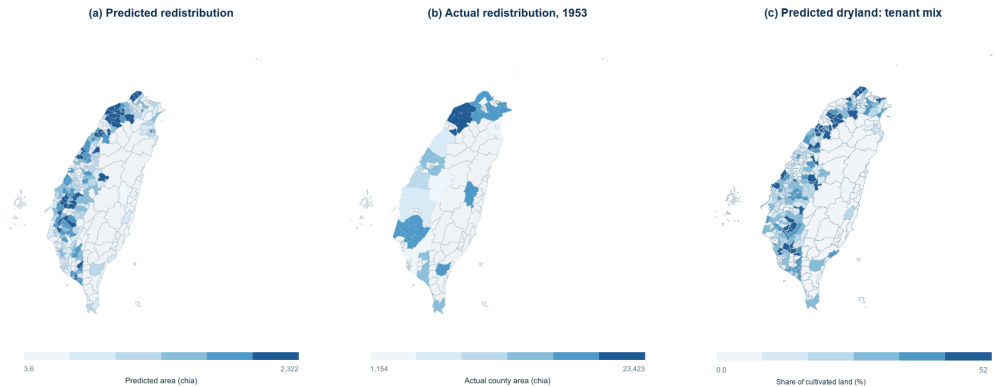
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Data sources and years

Source	Years
Colonial censuses and population statistics	1915–1935
County and municipal statistical yearbooks, statistical abstracts, household-registration records, and local gazetteers	1947–1954
《臺灣省戶籍統計要覽：民國三十五年至四十七年戶籍統計年報》	1955–1958
《耕地所有權與經營狀況調查報告書》	1950
Digitized research appendix by Chiang, Fan, and Hsu	1950, 1953
《農復會工作報告》第五期	1953
Taiwan Social Change Survey	2012, 2017, 2022
Manpower Utilization Survey	1978–2000
Internal Migration Survey	1980–1989, 1992
Central Election Commission	1998–2022

Predicted and actual redistribution across space



Research outcomes and land measures

Outcomes

- Registered township population
- Residence outside the birth township
- Employment composition and annual industry transitions
- Township mayoral elections

Land measures from the 1950 survey

$$DryShare_i = \frac{\text{township dryland}}{C_i}$$

Cash orientation of land reform: the township's dryland share identifies the form of compensation.

$$D_i = \underbrace{\frac{\widehat{R}_i}{C_i}}_{\text{predicted total redistribution}} \times \underbrace{\frac{Dry_i^{\text{tenant}}}{C_i^{\text{tenant}}}}_{\text{dryland share of tenant-operated land}}$$

Cash-compensation intensity: predicted redistributed dryland as a share of all township cultivated land.

$PublicShare_i =$ leased public cultivated land$/C_i$.

C_i: all township cultivated land. $\widehat{R}_i$: predicted private-land acquisition.

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Baseline continuous-treatment difference-in-differences

$$Population_{it} = \alpha_i + \lambda_t + \beta\big(DryShare_i \times Post_t\big) + \varepsilon_{it}$$

- For Land-to-the-Tiller, $Post_t = 1$ from 1952, when large-scale public communication began.
- Event studies omit 1951, the last year before the first treated year.
- Township and year fixed effects absorb local constants and Taiwan-wide shocks.
- Standard errors are clustered by the 237 treatment units.

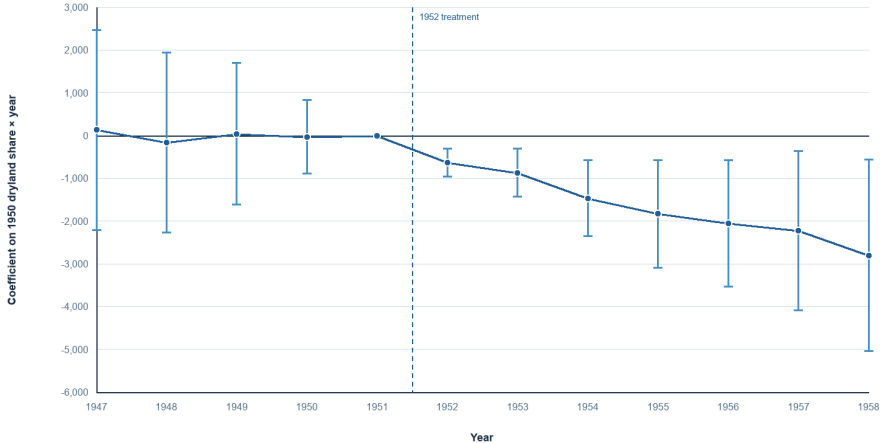
Year	1947	1948	1949	1950	1951	1952	1953	1954	1955	1956	1957	1958
Units	148	149	154	205	237	237	227	237	237	237	237	237

Baseline estimates and sample completeness

	All available (1)	95% coverage (2)	85% coverage (3)	Complete years (4)
$DryShare_i \times \mathbf{1}[t \geq 1952]$	-1697.273^{*}	-1702.579^{***}	-1692.290^{***}	-1836.304^{***}
	(867.396)	(622.278)	(650.008)	(685.846)
p-value	0.050	0.006	0.009	0.007
Township FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	2,542	1,886	2,091	1,659
Treatment units	237	237	237	237
Sample years	1947–1958	1951–1958	1950–1958	7 complete years

Dependent variable: population count. Coefficients and SE use the original 0–1 dryland share: one unit = 100 pp. For 10 pp, multiply both by 0.10 (column 1: -169.727 residents, SE 86.740). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Event-study estimates and pre-trend assessment



Error bars are unit-clustered 95% confidence intervals. Pre-treatment joint test for 1947–1950: $\chi^2(4) = 5.282$, $p = 0.260$. Post-treatment joint test for 1952–1958: $p < 0.001$.

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Construction of predicted redistribution

1. Fit the grouped owner-size distribution

$$(\widehat{\mu}_i, \widehat{\sigma}_i) = \arg\max_{\mu,\,\sigma>0} \sum_{b=1}^{12} \left(n_{ib} + 25\pi_{c(i)b}\right) \ln[F_i(u_b) - F_i(l_b)]\,.$$

$$F_i(a) = \Phi\left(\frac{\ln a - \mu_i}{\sigma_i}\right).$$

2. Predict land above the retention threshold

$$\widehat{R}_i = N_i\, \mathbb{E}_{\widehat{F}_i}[(A_i - h)_+] = N_i \int_h^{\infty} [1 - \widehat{F}_i(a)]\, da, \qquad h = 3 \text{ chia}.$$

3. Allocate the dryland component using the tenant land mix

$$R_i = \frac{\widehat{R}_i}{C_i}, \qquad D_i = R_i \times \frac{Dry_i^{\text{tenant}}}{C_i^{\text{tenant}}}.$$

n_{ib}: owners in size bin b, bounded by l_b, u_b. $\pi_{c(i)b}$: county bin share. N_i: township owner count.

Land-size bins are in hectares. The retention threshold is 3 chia = 2.909751 hectares; predicted area and township cultivated area are converted to the same unit.

The 1951 sale of public land



Institutional distinction

- The 1951 program covered government-held cultivated land.
- Incumbent cultivators purchased the land in installments.
- Its timing predates the 1953 acquisition of privately owned rented land.

Empirical proxy

Leased public cultivated land in the 1950 survey, measured in chia and transformed using a standardized inverse hyperbolic sine.

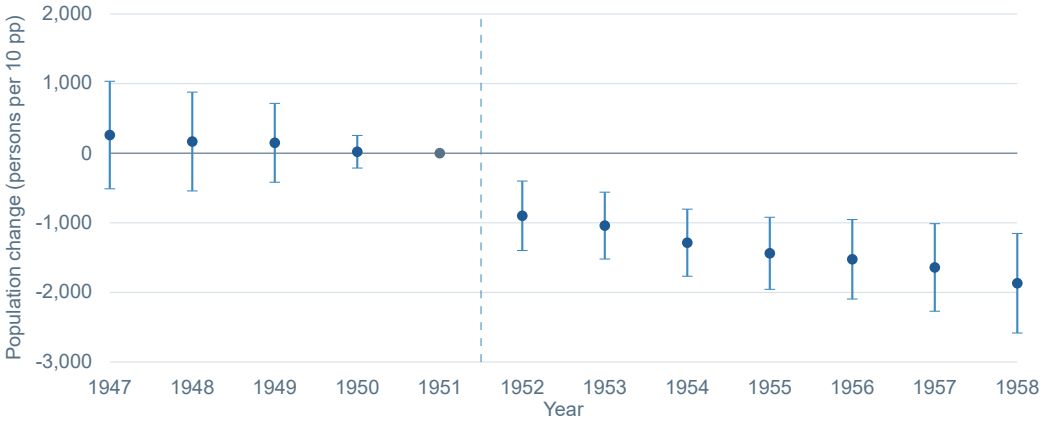
Population effects: predicted redistributed dryland

$$Population_{it} = \alpha_i + \lambda_t + \beta(D_i \times Post_t) + \gamma(PublicLease_i \times Post_t) + \delta(R_i \times Post_t) + \varepsilon_{it}.$$

	Predicted dryland	+ Public land	+ Both
$D_i \times Post_t$ (per 10 pp)	-778.784^{***} (269.855)	-795.396^{***} (271.324)	-1459.817^{***} (275.822)
$PublicLease_i \times Post_t$ (1 SD)	–	568.580^{***} (171.294)	656.889^{***} (152.882)
$R_i \times Post_t$ (per 10 pp)	–	–	525.489^{***} (163.070)
Township and year FE	Yes	Yes	Yes
Observations	2,542	2,542	2,542
Townships	237	237	237

$Post_t = 1$ from 1952. D_i, R_i are 0–1 shares. The table reports $0.10\widehat{\beta}$, $0.10\widehat{\delta}$ and their scaled SE, hence effects per 10 pp. Public land: per 1 SD. Township-clustered SE in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Event study with both reform exposures controlled



Pre-treatment joint test $\chi^2(4) = 1.957$, $p = 0.744$

Post-treatment joint test $\chi^2(7) = 44.593$, $p < 0.001$

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Migration and migrants' characteristics

Taiwan Social Change Survey, 2012, 2017, and 2022

Predicted redistributed dryland and migration

Sample: born before 1953 Outcome	D_i (per 10 pp)	Observations	Years
Residence outside the birth township (pp)	15.815^{**} (7.087)	692	2012–2022
Workplace outside the birth township (pp)	25.693^{***} (3.311)	193	2012–2022

Six historical controls; county, survey-year, and birth-year fixed effects.

Migrants compared with residents of their birth township

Sample: born before 1953	College attendance or above (pp)	Professional/technical occupation (pp)	Household income rank (percentile)	Financial products held/invested in (pp)
Migrant	10.52^{**} (4.37)	8.56^{**} (3.11)	13.78^{**} (6.02)	14.38^{**} (5.51)
Observations	690	664	426	532

Migrant: current township differs from birth township. Survey weights; year, birth-year, and birth-township fixed effects. Birth-county clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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Specification for long-run outcomes

$$y_{uit} = \alpha_{c(i)} + \lambda_t + \beta D_i^{(10)} + X_i'\gamma + \varepsilon_{uit}, \qquad D_i^{(10)} \equiv D_i/0.10.$$

1950/1951 controls	Log 1951 population, log 1950 cultivated area, share of owners with at least 3 ha, and tenant-operated land share
1961 controls	Log crop output value and double-cropped paddy share
Fixed effects	County and year. TSCS regressions also include birth-year fixed effects

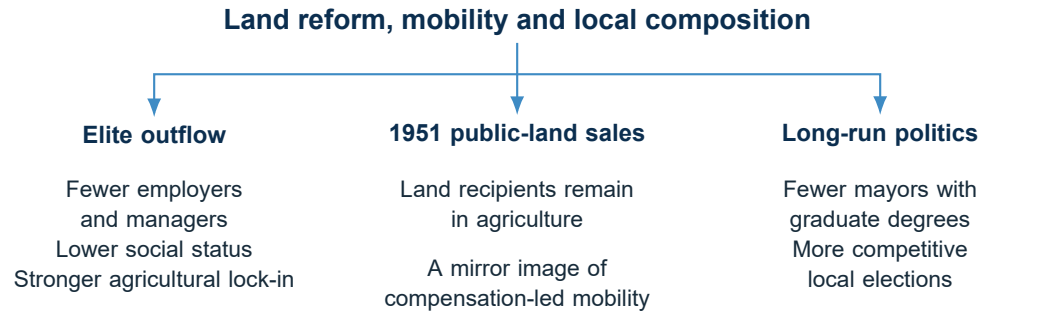
Why the 1961 variables proxy conditions at reform

The literature finds little effect of land reform on grain output (Kim and Wang, 2025). Cultivated land had largely undergone modernization under Japanese rule, with little subsequent change after reform.

The 1961 variables control for persistent **natural endowments and irrigation infrastructure** not fully captured in the 1950 survey.

Dropping both 1961 controls and retaining only the 1950/1951 baseline controls leaves the core results robust.

Conceptual roadmap: long-run consequences



The long-run evidence traces the departure of local elites, agricultural attachment among land recipients, and subsequent political change.

Agricultural lock-in and job mobility

Outcome (pp)	D_i (per 10 pp)	Observations	Years
A. Manpower Utilization Survey: linked annual records			
Agricultural-to-nonagricultural transition	-1.952^{*} (1.052)	30,886	1979–2000
Enter nonagricultural self-employment or employer status	-0.373^{**} (0.161)	86,028	1979–2000
Become an employer	-0.281^{*} (0.133)	123,292	1979–2000
Change work township in the following year	-1.042^{***} (0.270)	117,304	1979–2000
B. Manpower Utilization Survey: annual survey			
Want to change or add a job	-0.543^{**} (0.233)	311,246	1978–2000

Coefficients: per 10 pp of predicted redistributed dryland. County-clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Average social status

Taiwan Social Change Survey; born before 1953

Outcome	D_i (per 10 pp)	Observations	Years
Father attended junior college or above (pp)	-1.339^{*} (0.710)	587	2012, 2017, 2022
Self-rated social status (1–10)	-0.551^{**} (0.216)	581	2012, 2017, 2022
Political-group participation (pp)	-1.988^{*} (1.047)	541	2012, 2022

Coefficients: per 10 pp of predicted redistributed dryland. County-clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Human capital

Share reaching each educational level or above (pp)

Highest level attended	Manpower Utilization Survey	Internal Migration Survey	Taiwan Social Change Survey
Primary school or above	-0.693 (0.687)	-0.907 (0.874)	4.204 (3.228)
Junior high school or above	-1.473 (1.464)	-3.373 (2.705)	-9.388 (5.469)
Senior high / vocational school or above	-1.454 (1.626)	-3.387 (2.825)	-5.512 (5.167)
Junior college or above	0.049 (0.986)	-2.627 (1.883)	-3.606 (2.729)
Observations	311,653	215,195	690
Exposure geography	Workplace	Residence	Birthplace
Years	1978–2000	1980–1989, 1992	2012, 2017, 2022

Coefficients: per 10 pp of predicted redistributed dryland. County-clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Survey columns use sampling weights. TSCS: born before 1953. Educational levels refer to the highest level attended.

Basic education and elite mobility

Broad educational attainment

The education coefficients do not show a significant decline in attainment among the local population.

Postwar expansion of compulsory education sustained broad access to basic schooling.

The top of the local hierarchy

- **Employers:** fewer local entrepreneurs
- **Managers:** a thinner managerial stratum
- **Mayors with graduate degrees:** fewer highly educated political elites

Elite mobility concentrates losses at the top of the economic and political hierarchy, while basic education remains widespread.

Employment structure and working hours

Outcome	Manpower Utilization Survey	Internal Migration Survey
Employers (pp)	-0.833^{***}	-0.492^{***}
	(0.272)	(0.165)
Private employees (pp)	-4.574^{**}	-2.924
	(1.582)	(1.925)
Unpaid family workers (pp)	2.298^{**}	2.650^{*}
	(0.985)	(1.421)
Managerial occupations (pp)	-0.420^{**}	-0.201
	(0.142)	(0.163)
Agricultural employment (pp)	2.935	4.243^{**}
	(1.957)	(1.882)
Manufacturing employment (pp)	-3.927^{**}	-0.930
	(1.795)	(2.549)
Secondary-sector employment (pp)	-4.101^{**}	-1.653
	(1.857)	(2.830)
Weekly working hours (hours)	0.639	1.026^{**}
	(0.455)	(0.434)
Observations (shares / hours)	311,653	149,649 / 134,373
Years	1978–2000	1980–1989, 1992

Per 10 pp of predicted redistributed dryland. County-clustered SE in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.
Survey sampling weights; ages 25–64. IMS working hours: 1980–1989.

Public-land sales and employment

Outcome	Public-land share (per 10 pp)	Observations
Agricultural employment, pooled censuses	2.353^{**} (0.842)	1,135
Agricultural occupations	2.012^{**} (0.764)	908
Agricultural industry	2.999^{**} (1.244)	454
Agricultural-to-nonagricultural transition	-0.834^{**} (0.310)	30,886
Historical controls	Six	
County and year FE	Yes	

Land-share coefficients and SE: per 0.10 increase in the original 0–1 share (10 pp). Log controls: per log unit. Share outcomes: pp. Parentheses: historical-county CR1 clustered SE. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; two-sided $t(G-1)$ tests.

Township mayoral elections

Election sample

- Seven elections, 1998–2022
- 1,198 observations
- 232 historical townships in 15 counties

Outcomes

1. Mayor attended university or above
2. Mayor holds a graduate degree
3. Number of officially listed candidates
4. Winner’s share of valid candidate votes
5. Winner’s margin over the runner-up
6. Probability of a Kuomintang (KMT) victory

Township mayoral election outcomes

Predicted redistributed dryland; equal township-year weights

Outcome	D_i (per 10 pp)	Observations	Years
Mayor attended university or above (pp)	-7.984^{*} (4.125)	457	2009–2022
Mayor holds a graduate degree (pp)	-10.758^{***} (1.884)	346	2014–2022
Number of candidates	0.137^{**} (0.051)	1,198	1998–2022
Winner's vote share (pp)	-2.422^{***} (0.806)	1,198	1998–2022
Winner's margin over runner-up (pp)	-2.198^{***} (0.547)	1,093	1998–2022
Historical controls	Six		
County and year FE	Yes		

Land-share coefficients and SE: per 0.10 increase in the original 0–1 share (10 pp). Log controls: per log unit. Share outcomes: pp. Parentheses: historical-county CR1 clustered SE. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; two-sided $t(G-1)$ tests.

KMT support: elite outflow and land recipients

Land reform may affect KMT support through two channels: **elite outflow** and **gratitude toward the authorities or greater conservatism among land recipients**.

Channel	Explanatory variable	What it isolates
Elite outflow	$DryShare_i =$ township dryland$/C_i$	Cash orientation of compensation
Land recipients	$B_i = \dfrac{\text{1953 beneficiary households}}{\text{1951 agricultural households}}$	Exposure to receiving land

Predicted redistributed dryland combines compensation type with the **extent of redistribution**. Township dryland share separates the cash channel from redistribution intensity.

The next table contrasts these channels: township dryland share is significant, while beneficiary household share is not, consistent with the elite-mobility mechanism.

KMT victories: a horse race

$$KMT_{it} = \alpha_{c(i)} + \lambda_t + \beta DryShare_i^{(10)} + \theta B_i^{(10)} + X_i'\gamma + \varepsilon_{it}.$$

	(1) Township dryland	(2) Beneficiary households	(3) Horse race
Township dryland share (per 10 pp)	2.293^{**} (0.981)	–	2.708^{**} (1.004)
Beneficiary household share (per 10 pp)	–	1.169 (1.755)	2.331 (1.721)
Historical controls	Six	Six	Six
County and election-year FE	Yes	Yes	Yes
Observations	1,152	1,152	1,152
Historical townships	222	222	222
County clusters	15	15	15
R^2	0.184	0.179	0.187

Outcome: KMT victory (pp). All columns use the same complete sample. $DryShare_i^{(10)} = DryShare_i/0.10$ and $B_i^{(10)} = B_i/0.10$. Beneficiary share: 1953 beneficiary households / 1951 agricultural households.

Historical-county CR1 clustered SE in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; two-sided $t(14)$ tests. Horse-race p-values: township dryland 0.017; beneficiary households 0.197.

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- **Mobility:** Predicted redistributed dryland increases residence and employment outside the birth township. Migrants have higher education, income rank, and financial-asset participation.
- **Local economy:** Predicted redistributed dryland reduces employers, managers, and entry into nonagricultural self-employment.
- **Local politics:** Predicted redistributed dryland increases the number of candidates and lowers winners' vote shares and victory margins.
- **Public-land sales:** Greater public-land exposure reduces agricultural-to-nonagricultural transitions.

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Outline

Historical Background and Literature Review

Data and Measurement

Short-run baseline population results

Estimating and controlling confounding factors from the land reform

Individual-Level Evidence on Migration

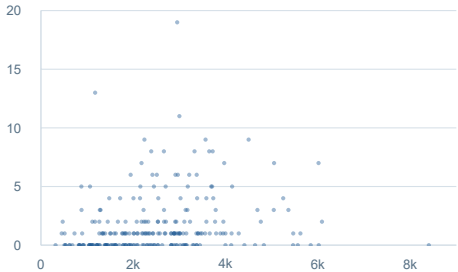
Long-run Outcomes

Conclusion

Appendix

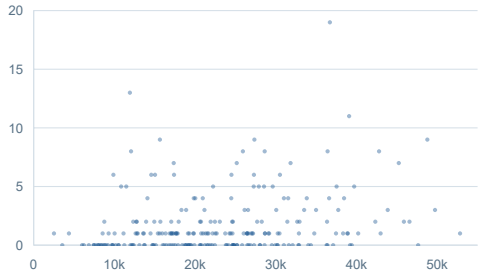
Large landowners and township size

1950 cultivated land (chia)



$r = 0.199$

1951 population (persons)



$r = 0.245$

Owner counts rise only modestly with township size. Population levels retain the direct interpretation in persons.

$N = 237$. Vertical axis: private-land owners with at least 30 hectares (about 30.9 chia). The 1950 survey reports hectares in bins; r denotes the Pearson correlation.

Step 1: observed inputs from the 1950 land survey

Private cultivated-land owners by holding-size bin

For township i, the survey reports n_{ib} owners in each of 12 ownership-area bins b:

$[0, 0.5)$, $[0.5, 1)$, $[1, 2)$, $[2, 3)$, $[3, 5)$, $[5, 7)$ ha
$[7, 10)$, $[10, 20)$, $[20, 30)$, $[30, 50)$, $[50, 100)$, $[100, \infty)$ ha

$$N_i = \sum_{b=1}^{12} n_{ib}$$

is the number of owners represented in the grouped distribution.

Other township-level fields

- total cultivated area C_i
- paddy area and dryland area Dry_i
- cultivated area by operating status, including tenant-operated paddy and dryland

Original source title: 《耕地所有權與經營狀況調查報告書》. The owner-size bins are measured in hectares; later calculations convert hectares and chia using $1 \text{ chia} = 0.969917$ hectares.

Step 2: fit a township owner-size distribution

Recommended distribution

$$F_i(a) = \Pr(A_i \le a) = \Phi\left(\frac{\ln a - \mu_i}{\sigma_i}\right)$$

County-bin shares $\pi_{c(i)b}$ supply limited shrinkage for small townships:

$$\widetilde{n}_{ib} = n_{ib} + \kappa \pi_{c(i)b}, \qquad \kappa = 25.$$

Parameters are estimated from the grouped likelihood:

$$(\widehat{\mu}_i, \widehat{\sigma}_i) = \arg\max_{\mu,\sigma} \sum_b \widetilde{n}_{ib} \ln[F_i(u_b) - F_i(l_b)] \,.$$

What the shrinkage does

- Adds 25 pseudo-owners distributed according to the 1950 county bin shares.
- Stabilizes the bin proportions of townships with few observed owners.
- Uses the 1950 land survey; N_i and fitted excess area determine each township estimate.

Model comparison

Five candidates are compared: lognormal, log-logistic, gamma, generalized gamma, and lognormal with a Pareto tail. The selected specification is lognormal with $\kappa = 25$.

The township likelihood uses the 1950 survey inputs. The 1953 county totals enter model comparison and validation.

Step 3: construct treatment intensity

Expected land above a uniform threshold

Let $h = 3$ chia (2.909751 hectares). For one owner,

$$e_i = \mathbb{E}[(A_i - h)_+] = \int_h^\infty [1 - F_i(a)]\, da.$$

For a lognormal distribution,

$$e_i = \exp\left(\widehat{\mu}_i + \frac{\widehat{\sigma}_i^2}{2}\right) \Phi\left(\frac{\widehat{\mu}_i + \widehat{\sigma}_i^2 - \ln h}{\widehat{\sigma}_i}\right) - h\Phi\left(\frac{\widehat{\mu}_i - \ln h}{\widehat{\sigma}_i}\right).$$

Predicted township redistribution is

$$\widehat{R}_i = N_i e_i,$$

after converting the result back from hectares to chia.

Dryland component and normalization

$$q_i = \frac{Dry_i^{\text{tenant}}}{C_i^{\text{tenant}}}, \qquad \widehat{R}_i^D = \widehat{R}_i q_i,$$

$$D_i = \frac{\widehat{R}_i^D}{C_i}.$$

Assume redistributed land has the same dryland share as tenant-operated cultivated land in 1950.

D_i is the regression treatment: predicted redistributed dryland divided by township cultivated land.

Validation against 1953 county totals

Construct	Predicted area	Actual area	Predicted / actual	Level corr.	Log corr.
Total redistributed land	122,313	139,003	0.880	0.852	0.923
Dryland: tenant land mix	30,204	21,385	1.412	0.800	0.886

